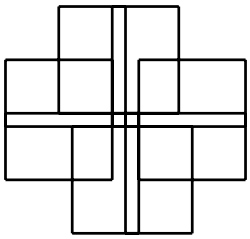
LESSON 12
SQUARING

SUMMARY



12.1 Squaring Numbers that end in 5
12.2 Squaring Numbers Near 50
12.3 General Squaring – from left to right.
12.4 Number Splitting – to simplify squaring calculations.
12.5 Algebraic Squaring
12.6 Digit Sums of squares – properties of square numbers.
12.7 Square Roots of Perfect Squares – where the answer is a 2-figure number.
12.8 3 and 4-Figure Numbers – squaring bigger numbers.

12.1 SQUARING NUMBERS THAT END IN 5

Squaring is multiplication in which a number is multiplied by itself:
so 75×75 is called "75 squared" and is written 75^2.

The formula *By One More Than the One Before* provides a beautifully simple way of squaring numbers that end in 5.

1. In the case of $\mathbf{75^2}$, we simply multiply the 7 (the number before the 5) by the next number up, 8. This gives us 56 as the first part of the answer, and the last part is simply 25 (5^2).

   So $\mathbf{75^2 = 56/25}$ where $56=7\times8$, $25=5^2$.

2. Similarly $\mathbf{65^2 = 4225}$ $42=6\times7$, $25=5^2$.

3. And $\mathbf{25^2 = 625}$ where $6=2\times3$.

4. Also since 4½= 4.5, the same method applies to squaring numbers ending in ½.
   So $\mathbf{4½^2 = 20¼}$, where $20 = 4\times5$ and $¼ = ½^2$.

   The method can be applied to numbers of any size:

5. $\mathbf{305^2 = 93025}$ where $930 = 30\times31$.

Even for large numbers like, say, 635, it is still easier to multiply 63 by 64 and put 25 on the end than to multiply 635 by 635.

Algebraic Proof: $(ax + 5)^2 = a(a + 1)x^2 + 25$, where x = 10. See also end of section 11.2.

✐ **Practice A** Square the following numbers:

a 55	**b** 15	**c** 8½	**d** 95
e 105	**f** 195	**g** 155	**h** 245
i 35	**j** 20½	**k** 8005	**l** 350

What number, when squared, gives:

m 2025 **n** 30¼ **o** 902500

a 3025	**b 225**	**c 72¼**	**d 9025**
e 11025	**f 38025**	**g 24025**	**h 60025**
i 1225	**j 420¼**	**k 64080025**	**l 122500**
m 45	**n 5½**	**o 950**	

12.2 SQUARING NUMBERS NEAR 50

Here is another special squaring method.

6 **53^2 = 2809**.

The answer is in two parts: 28 and 09.
28 is simply the last figure, 3, increased by 25.
And 09 is just 3^2.

7 Similarly **52^2 = 2704** ($2 = 2 + 25$, $04 = 2^2$).

Algebraic Proof: $(50 + a)^2 = 100(25 + a) + a^2$.

✐ **Practice B** Find:

a 54^2 **b** 56^2 **c** 57^2 **d** 58^2 **e** 61^2

f 62^2 **g** 51^2

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a 2916 b 3136 c 3249 d 3364 e 3721
f 3844 g 2601

8 **47^2 = 2209**.

Similarly, for numbers below 50 we take the deficiency from 50 (3 here) **from** 50, to get 47 in this case, and put the square of the deficiency, **9**.

In the proof above 'a' would take negative values for numbers below 50.

✐ **Practice C** Square the following numbers by this method:

a 46 **b** 44 **c** 42 **d** 39 **e** 43

f 49 **g** 41 **h** 37

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a 2116 b 1936 c 1764 d 1521 e 1849
f 2401 g 1681 h 1369

12.3 GENERAL SQUARING

The *Vertically and Crosswise* formula simplifies nicely when the numbers being multiplied are the same, and gives us a very easy method for squaring numbers.

THE DUPLEX

We will use the term **Duplex**, D, as follows:

for 1 figure **D is its square**, e.g. D(4) = 4^2 = **16**;

for 2 figures **D is twice their product**, e.g. D(43) = 2×4×3 = **24**.

✎ **Practice D** Find the Duplex of the following numbers:

a 5	**b** 23	**c** 55	**d** 2
e 14	**f** 77	**g** 26	**h** 90

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a 25	**b 12**	**c 50**	**d 4**
e 8	**f 98**	**g 24**	**h 0**

The square of any number is just the total of its Duplexes, combined in the way we have been using for mental multiplication.

9 **43^2 = 1849**.

Working from left to right there are three duplexes in 43:D(4), D(43) and D(3).

D(4) = **16**, D(43) = **24**, D(3) = **9**,

combining these three results in the usual way we get

16
16,24 = 184
184,9 = **1849**.

10 **64^2 = 4096**.

D(6) = **36**, D(64) = **48**, D(4) = **16**,

So mentally we get

36
36,48 = 408
408,16 = **4096**.

Algebraic proof: $(10a + b)^2 = 100(a^2) + 10(2ab) + b^2$. This method can also be explained by multiplying a number by itself using the general multiplication method.

✎ **Practice E** Square the following:

a 31	**b** 14	**c** 41	**d** 26
e 23	**f** 32	**g** 21	**h** 66
i 81	**j** 91	**k** 56	**l** 63
m 77	**n** 33		

a	**961**	**b**	**196**	**c**	**1681**	**d**	**676**
e	**529**	**f**	**1024**	**g**	**441**	**h**	**4356**
i	**6561**	**j**	**8281**	**k**	**3136**	**l**	**3969**
m	**5929**	**n**	**1089**				

Duplexes and squares of longer numbers are covered in Section 12.8

12.4 NUMBER SPLITTING

You may recall that we could sometimes group two figures as one when we were multiplying two 2-figure numbers together (see Section 11.4). This also applies to squaring.

11 **123^2 = 15129**.

Here we may think of 123 as 12/3, as if it were a 2-figure number:

D(12) = 12^2 = **144**,
D(12/3) = 2 × 12 × 3 = **72**,
D(3) = 3^2 = **9**.

Combining these: 14 4,7 2 = 1512, and 1512,9 = **15129**.

✐ **Practice F** Square the following, grouping the first pair of figures together:

a	121	**b**	104	**c**	203	**d**	113
e	116	**f**	108	**g**	111		

a	**14 641**	**b**	**10 816**	**c**	**41 209**	**d**	**12 769**	**e**	**13 456**	**f**	**11 664**	**g**	**12321**

The other way of splitting the numbers, shown in Section 11.4 can also be used here.

12 **312^2 = 97344**.

Here we can split the number into 3/12 but we must work with **pairs of digits**:

D(3) = **9**, D(3/12) = **72**, D(12) = **144**.

Combining: 9,72 = 972 we can put both figures of 72 after the 9,
97 2,144 = **97344**.

✐ **Practice G** Square the following, grouping the last 2 figures together:

a 211 **b** 412 **c** 304 **d** 902

e 407 **f** 222 **g** 711

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a 44 521 b 169 744 c 92 416 d 813 604
e 165 649 f 49 284 g 505 521

12.5 ALGEBRAIC SQUARING

Exactly the same method we have been using for squaring numbers can be used for squaring algebraic expressions.

13 Find $\mathbf{(x + 5)^2}$.

This is just like squaring numbers: we find the duplexes of x, x+5 and 5.

D(x) = $\mathbf{x^2}$, D(x+5) = 2×x×5 = **10x**, D(5) = 5^2 = **25**.

So $\mathbf{(x + 5)^2 = x^2 + 10x + 25}$.

14 Find $\mathbf{(2x + 3)^2}$.

There are three Duplexes: D(2x) = $\mathbf{4x^2}$, D(2x+3) = 2×2x×3 = **12x**, D(3) = **9**.

So $\mathbf{(2x + 3)^2 = 4x^2 + 12x + 9}$.

15 Find $\mathbf{(x – 3y)^2}$.

Similarly: D(x) = $\mathbf{x^2}$, D(x–3y) = 2×x×–3y = **–6xy**, D(–3y) = $\mathbf{9y^2}$.

So $\mathbf{(x – 3y)^2 = x^2 – 6xy + 9y^2}$.

✐ **Practice H** Square the following:

a (3x + 4) **b** (5y + 2) **c** (2x – 1) **d** (x + 7)

e (x – 5) **f** (x + 2y) **g** (3x + 5y) **h** (2a + b)

i (2x – 3y) **j** (x + y) **k** (x – y) **l** (x – 8y)

a $9x^2+24x+16$	**b** $25y^2+20y+4$	**c** $4x^2-4x+1$	**d** $x^2+14x+49$
e $x^2-10x+25$	**f** $x^2+4xy+4y^2$	**g** $9x^2+30xy+25y^2$	**h** $4a^2+4ab+b^2$
i $4x^2-12xy+9y^2$	**j** $x^2+2xy+y^2$	**k** $x^2-2xy+y^2$	**l** $x^2-16xy+64y^2$

12.6 DIGIT SUMS OF SQUARES

Investigations of square numbers can make interesting and useful lessons, leading for example to the following results.

Square numbers only have digit sums of 1, 4, 7, 9
and they only end in 1, 4, 5, 6, 9, 0.

This means that square numbers cannot have certain digit sums and they cannot end with certain figures.

In the exercise below some numbers cannot be square numbers according to the above results.

✐ **Practice I** Which are not square numbers (judging by the above results)?

a 4539	**b** 5776	**c** 6889	**d** 5271
e 104976	**f** 65436	**g** 27478	**h** 75379

a, d, f, g

If a number has a valid digit sum and a valid last figure that does not mean that it is a square number. The last number in the exercise, 75379, is not a square number even though it has an allowed digit sum of 4 and an allowed last figure of 9.

12.7 SQUARE ROOTS OF PERFECT SQUARES

16 Find $\sqrt{\mathbf{6889}}$.

First note that there are two groups of figures, 68'89, so we expect a 2-figure answer.

Next we use *The First by the First and the Last by the Last.* Looking at the 68 at the beginning we can see that since 68 is greater than 64 (8^2) and less than 81 (9^2) the first figure must be 8.

Or looking at it another way 6889 is between 6400 and 8100

$$\begin{aligned} 6400 &= 80^2 \\ \mathbf{6889} &= \mathbf{8?^2} \\ 8100 &= 90^2 \end{aligned}$$

so $\sqrt{6889}$ must be between 80 and 90. I.e. it must be eighty something.

Now we look at the last figure of 6889, which is 9.
Any number ending with 3 will end with 9 when it is squared so the number we are looking for could be 83.

But any number ending in 7 will also end in 9 when it is squared so the number could also be 87.

So is the answer 83 or 87?

There are two easy ways of deciding. One is to use the digit sums.
If $87^2 = 6889$ then converting to digit sums we get $6^2 \rightarrow 4$, which is not correct.
But $83^2 = 6889$ becomes $2^2 \rightarrow 4$, so the answer must be **83**.

The other method is to recall that since $85^2 = 7225$ and 6889 is **below** this $\sqrt{6889}$ must be **below 85**. So it must be **83**.

To find the square root of a perfect 4-digit square number
we find the first figure by looking at the first figures
and we find two possible last figures by looking at the last figure.
We then decide which is correct either by considering the digit sums
or by considering the square of their mean.

17 Find $\sqrt{\mathbf{5776}}$.

The 57 at the beginning is between 49 and 64, so the first figure must be 7.

The 6 at the end tells us the square root ends in 4 or 6.
So the answer is 74 or 76.

$74^2 = 5776$ becomes $2^2 \rightarrow 7$ which is not true in terms of digit sums, so 74 is not the answer.
$76^2 = 5776$ becomes $4^2 \rightarrow 7$, which is true, so **76** is the answer.

Alternatively to choose between 74 and 76 we note that $75^2 = 5625$ and 5776 is greater than this so the square root must be greater than 75. So it must be **76**.

In the following exercise try to find the answers mentally if you can, writing down only the answers.

✐ **Practice J** Find the square root of:

a 2116	**b** 5329	**c** 1444	**d** 6724
e 3481	**f** 4489	**g** 8836	**h** 361
i 784	**j** 3721	**k** 2209	**l** 4225
m 9604	**n** 5929		

a 46	**b 73**	**c 38**	**d 82**
e 59	**f 67**	**g 94**	**h 19**
i 28	**j 61**	**k 47**	**l 65**
m 98	**n 77**		

As you will have seen, square numbers ending in 5 must have a square root ending in 5, there is only one possibility for the last figure.

12.8 3 AND 4-FIGURE NUMBERS

This follows on from Section 12.3.
As shown before, the duplex of a 1-digit number is its square: e.g. D(4) = 4^2 = 16.
And the duplex of a 2-digit number is twice the product of the digits: e.g. D(35) = 2×3×5 = 30.

We can also find the duplex of 3-digit numbers or bigger.

For 3 digits D is **twice the product of the outer pair + the square of the middle digit**, e.g. D(137) = 2×1×7 + 3^2 = **23**;

for 4 digits D is **twice the product of the outer pair + twice the product of the inner pair**, e.g. D(1034) = 2×1×4 + 2×0×3 = **8**;

D(10345) = 2×1×5 + 2×0×4 + 3^2 = **19**;

and so on.

✐ **Practice K** Find the duplex of the following numbers:

a 3	**b** 34	**c** 47	**d** 1	**e** 88
f 234	**g** 282	**h** 111	**i** 304	**j** 270
k 1234	**l** 3032	**m** 7130	**n** 20121	**o** 32104

a 9	**b 24**	**c 56**	**d 1**	**e 128**
f 25	**g 72**	**h 3**	**i 24**	**j 49**
k 20	**l 12**	**m 6**	**n 5**	**o 25**

As with 2-figure numbers the square of a number is just the total of its duplexes.

18 **341^2 = 116281**.

Here we have a 3-figure number:

D(3) = 9, D(34) = 24, D(341) = 22, D(41) = 8, D(1) = 1.

Mentally: 9,24 = 114
∪
114,22 = 1162
∪
1162,8,1 = **116281**.

19 **4332^2 = 18766224**.

D(4) = 16, D(43) = 24, D(433) = 33, D(4332) = 34,
D(332) = 21, D(32) = 12, D(2) = 4.

Mentally:

16,24 = 184

184,33 = 1873

1873,34 = 18764

18764,21 = 187661

187661,12 = 1876622

1876622,4 = **18766224**.

Practice L Square the following numbers:

a 212	**b** 131	**c** 204	**d** 513
e 263	**f** 264	**g** 313	**h** 217
i 3103	**j** 2132	**k** 1414	**l** 4144

a 44 944	**b 17 161**	**c 41 616**	**d 263 169**
e 69 169	**f 69 696**	**g 97 969**	**h 47 089**
i 9 628 609	**j 4 545 424**	**k 1 999 396**	**l 17 172 736**

"whatever is consistent with right reasoning should be accepted, even though it comes from a boy or even from a parrot; and whatever is inconsistent therewith ought to be rejected, although emanating from an old man or even from the great sage Shree Shuka himself".
quoted in "Vedic Mathematics", Page 1d.